Digestion & Absorption of carbohydrates

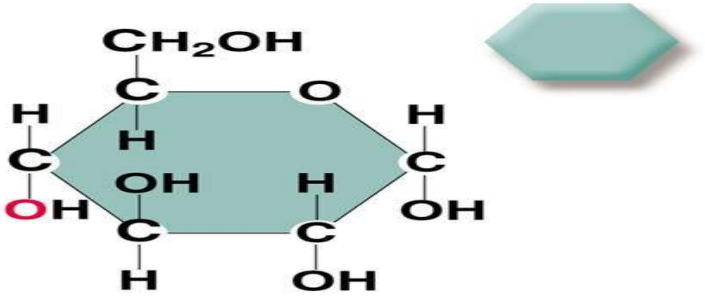
Learning Objectives

- At the end of lecture, student will be able to understand the basic concepts relevant to absorption and digestion of Carbohydrates

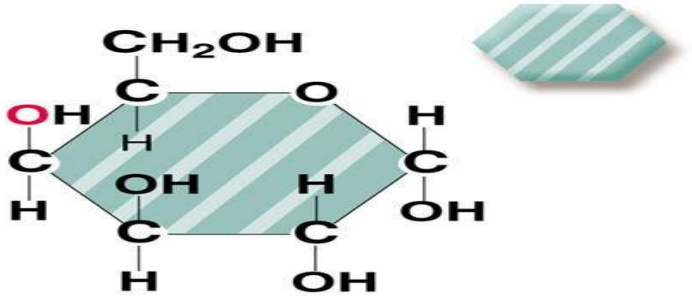
- Digestion is a process involving the hydrolysis of large and complex organic molecules of foodstuffs into smaller and preferably water-soluble molecules which can be easily absorbed by the GIT for utilization by the organism
- Digestion of macromolecules also promotes the absorption of fat soluble vitamins and certain minerals

- The principal dietary carbohydrates are polysaccharides (starch, glycogen) disaccharides (lactose, sucrose) & monosaccharides (glucose, fructose)
- The digestion of carbohydrates occurs in the mouth & intestine
- The hydrolysis of glycosidic bonds is carried out by a group of enzymes called glycosidases

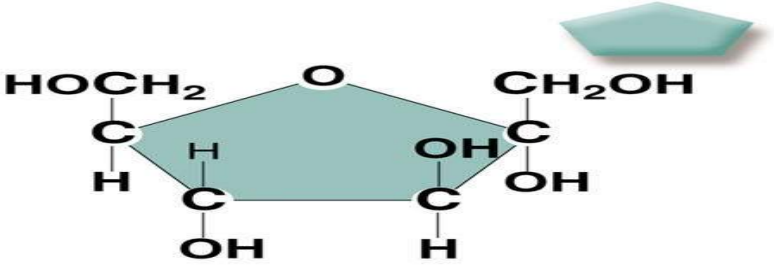
Monosaccharides



Glucose
(basic unit of polysaccharides)



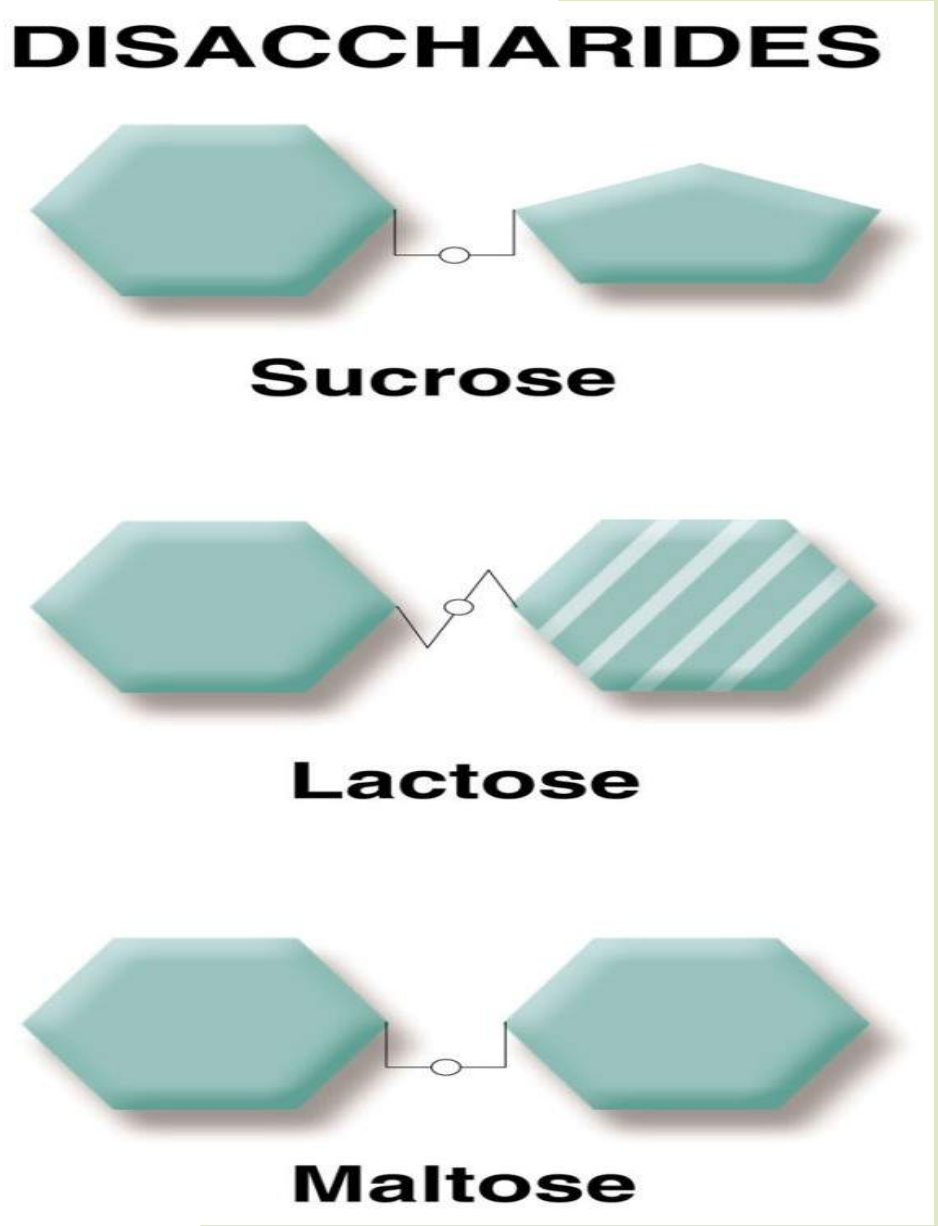
Galactose
(found as part of lactose in milk)



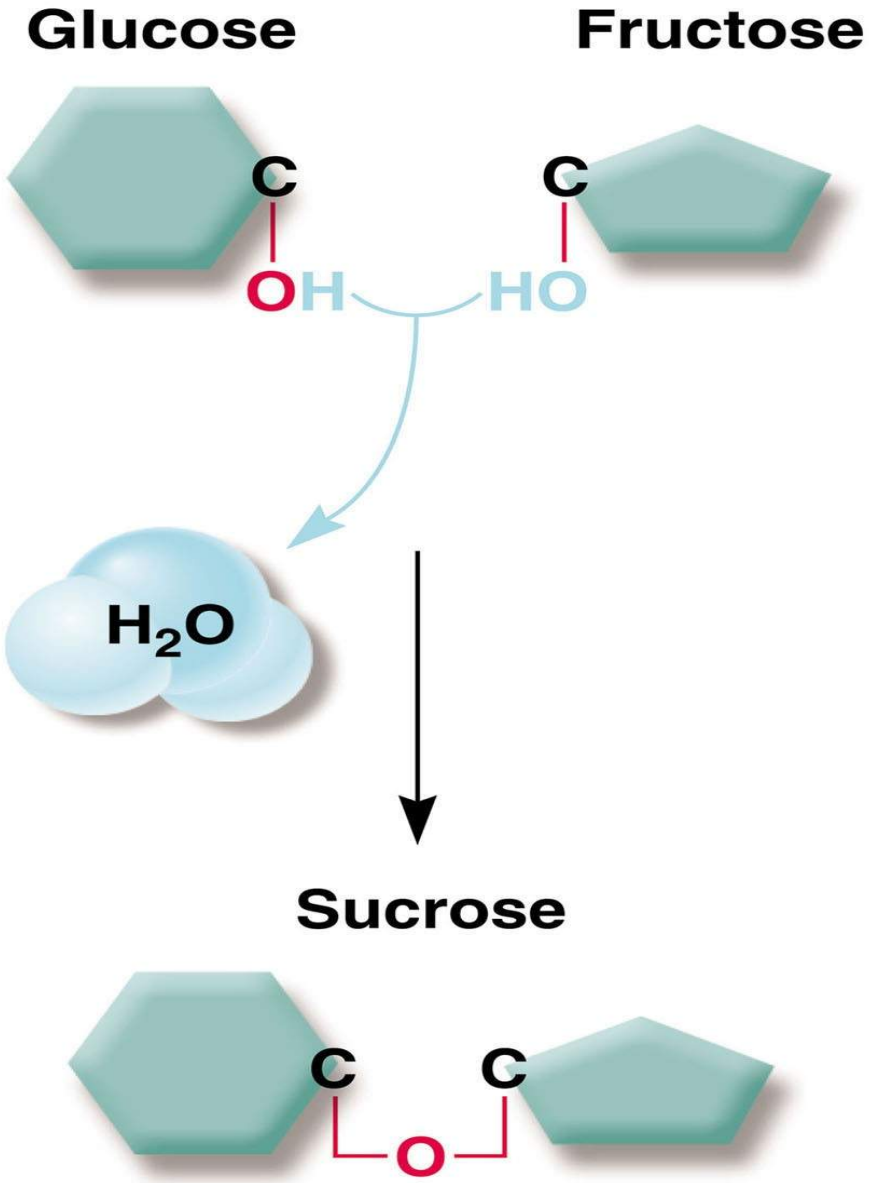
Fructose
(found in fruits, vegetables and honey)

DISACCHARIDES

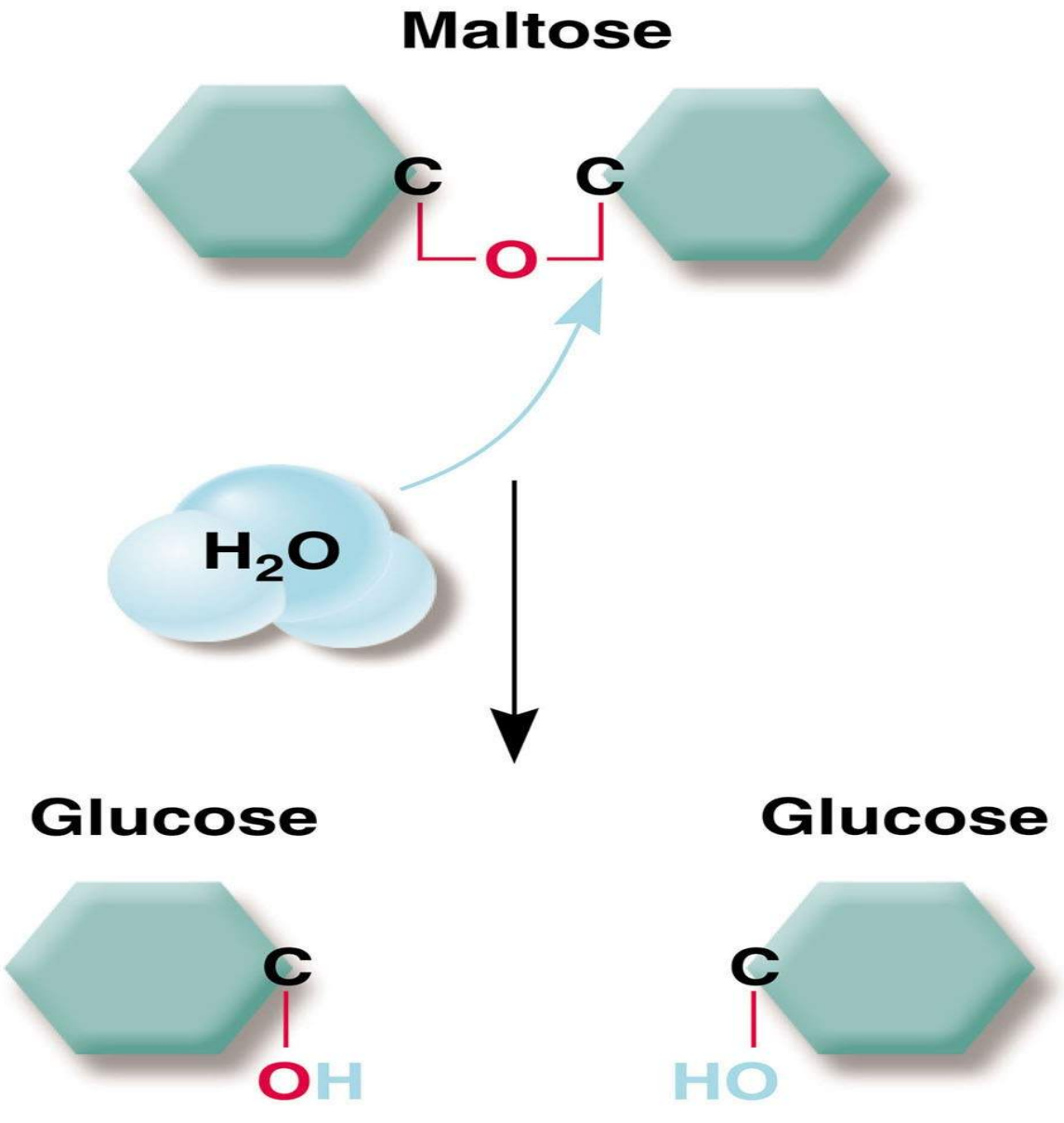
- Sucrose (glucose+fructose)
- Lactose (glucose+galactose)
- Maltose (glucose+glucose)



CONDENSATION

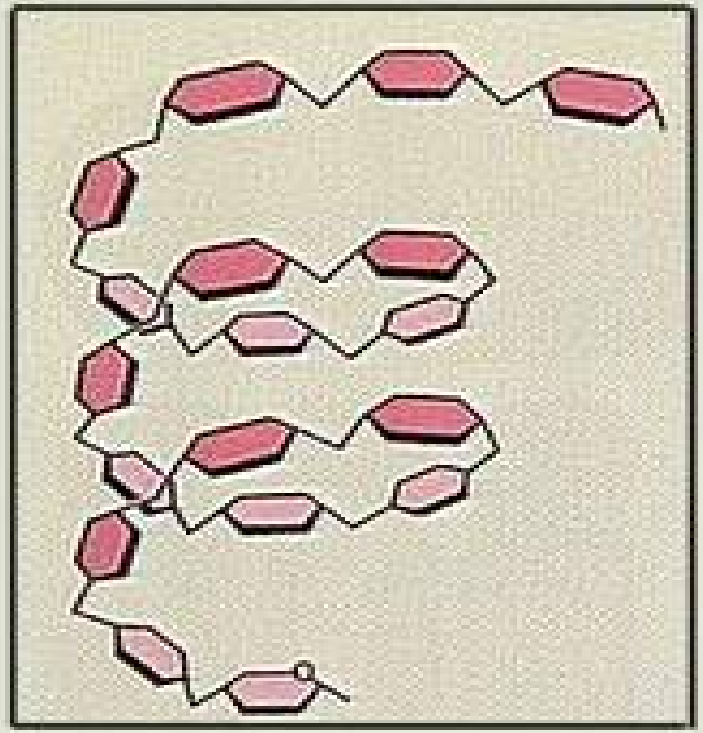


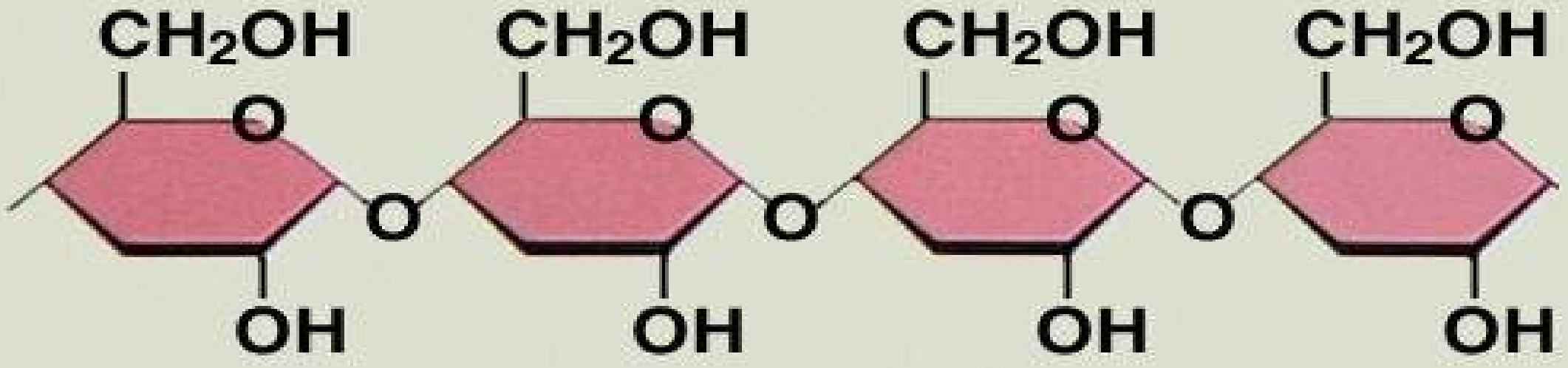
HYDROLYSIS



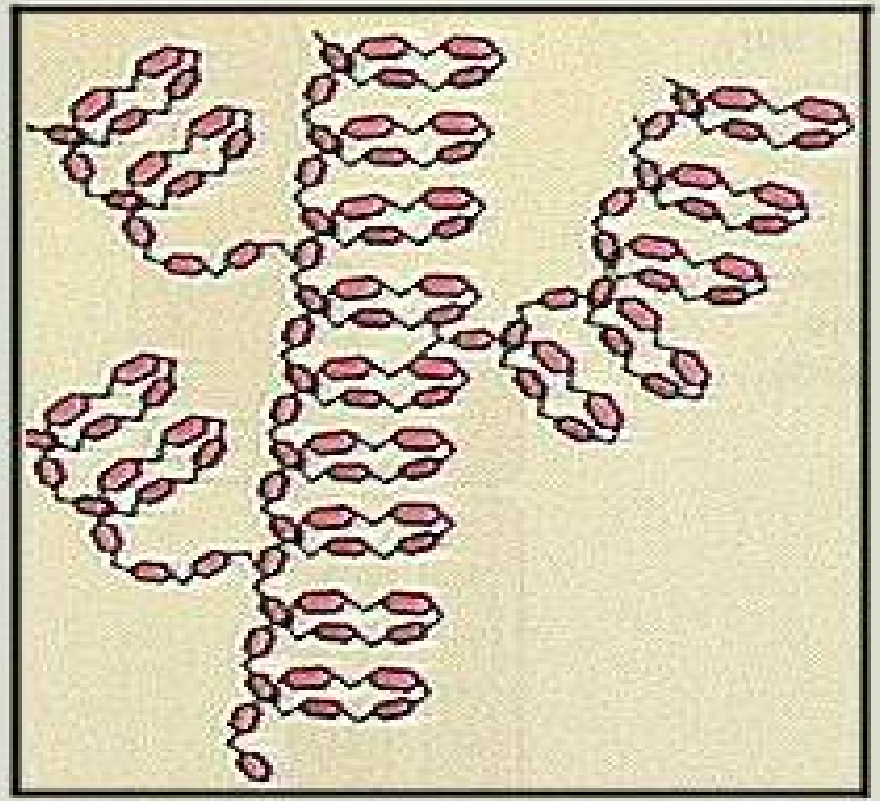
Amylose

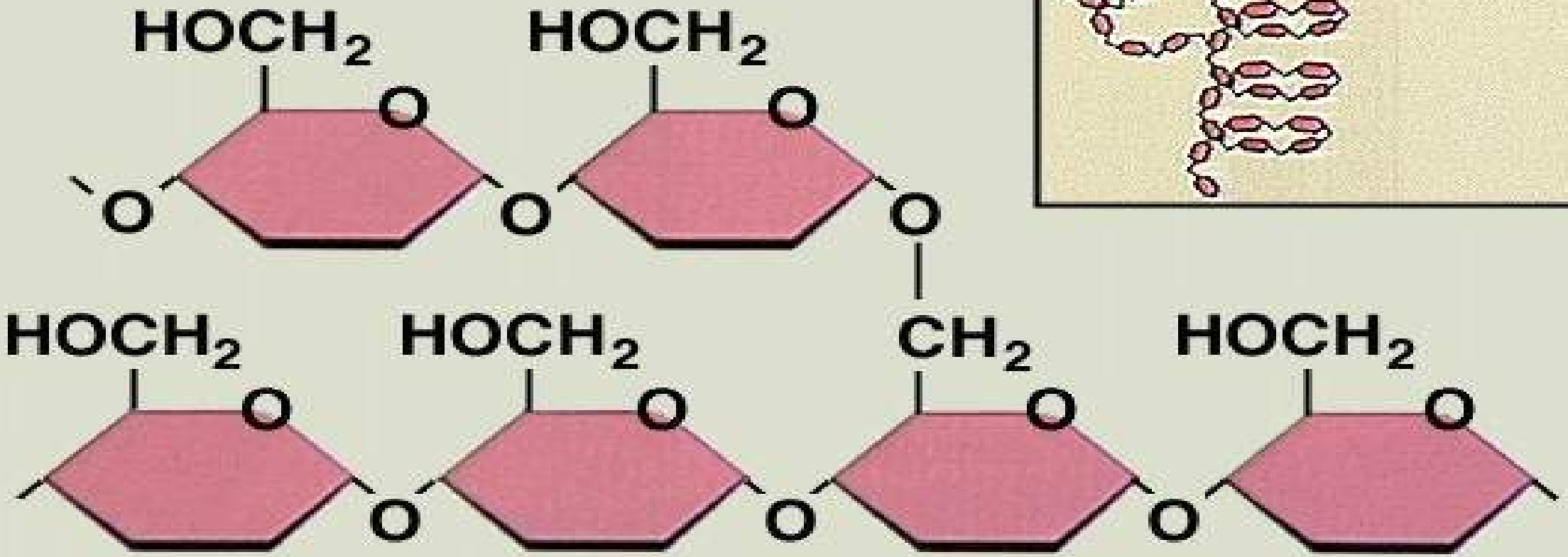
(starch)





Glycogen





Digestion in mouth:

- Saliva contains carbohydrate splitting enzyme salivary amylase (ptyalin)
- Action of salivary amylase (ptyalin):
- It is α – amylase, requires Cl^- ions for activation & optimum pH 6.7 (6.6 to 6.8)
- Salivary amylase hydrolyses α 1-4 glycosidic bonds of polysaccharides, producing smaller molecules maltose, glucose & trisaccharide, maltotriose

Salivary amylase action stops in stomach when pH falls to 3.0

- Digestion in stomach:
- No carbohydrate splitting enzyme in gastric juice
- Some dietary sucrose may be hydrolysed to equimolar amounts of glucose & fructose by HCL
- Digestion in duodenum:
- Food bolus in duodenum mixes with pancreatic juice
- Pancreatic juice contains pancreatic amylase, similar to salivary amylase

Action of pancreatic amylase:

- It is an α-amylase, optimum pH 7.1, requires Cl^- ions
- It specifically hydrolyzes α1-4 glycosidic bonds & not on a1-6 bonds
- It produces disaccharides (maltose, isomaltose) & oligosaccharides
- The final digestion of di- & oligosaccharides to monosaccharides primarily occurs at the mucosal lining of the upper jejunum

Carried out by oligosaccharidases (e.g. glucoamylase acting on amylose) and disaccharidases (e.g. maltase, sucrase, lactase)

- Digestion in small intestine:
- Action of intestinal juice:
- Intestinal amylase: It hydrolyses terminal a 1-4-glycosidic bonds in polysaccharides & oligosaccharides, liberating free glucose
- Lactase: It is β-galactosidase, its pH range 5.4 to 6.0
- Lactose is hydrolysed to glucose & galactose

- Isomaltase:
- It catalyses a 1-6 glycosidic bonds, branching points, producing maltose & glucose
- Maltase:
- It hydrolyses a 1-4-glycosidic bonds between glucose units in maltose & its pH range is 5.8 to 6.2
- Sucrase:
- It hydrolyses sucrose to glucose & fructose
- Its pH range is 5.0 to 7.0

Starch, glycogen & dextrins $\xrightarrow{\alpha\text{ - Amylase}}$ Glucose, maltose, isomaltose

Lactose $\xrightarrow{\text{Lactase}}$ Glucose + galactose

Maltose $\xrightarrow{\text{Maltase}}$ Glucose + Glucose

Sucrose $\xrightarrow{\text{Sucrase}}$ Glucose + Fructose

STARCH

Mouth and salivary glands
The salivary glands secrete saliva into the mouth to moisten the food. The salivary enzyme amylase begins digestion:

Starch —(Amylase)→ Small polysaccharides, maltose

Stomach
Stomach acid inactivates salivary enzymes, halting starch digestion.

Small intestine and pancreas
The pancreas produces an amylase that is released through the pancreatic duct into the small intestine:

Starch —(Pancreatic amylase)→ Small polysaccharides, maltose

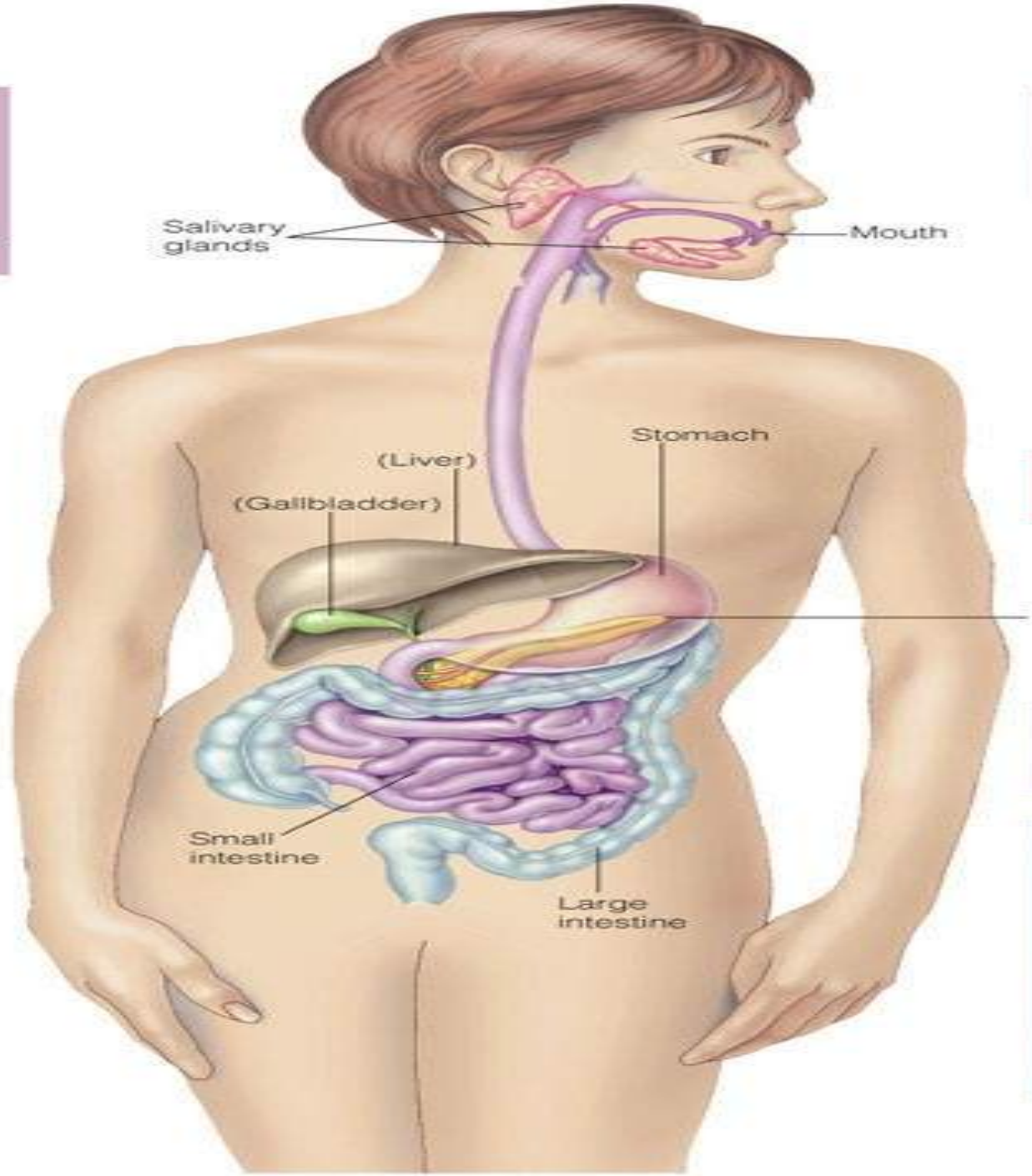
Then disaccharidase enzymes on the surface of the small intestinal cells hydrolyze the disaccharides into monosaccharides:

Maltose —(Maltase)→ Glucose + Glucose

Sucrose —(Sucrase)→ Fructose + Glucose

Lactose —(Lactase)→ Galactose + Glucose

Intestinal cells absorb these monosaccharides.



FIBER

Mouth
The mechanical action of the mouth crushes and tears fiber in food and mixes it with saliva to moisten it for swallowing.

Stomach
Fiber is not digested, and it delays gastric emptying.

Small intestine
Fiber is not digested, and it delays absorption of other nutrients.

Large intestine
Most fiber passes intact through the digestive tract to the large intestine. Here, bacterial enzymes digest fiber:

Some fiber —(Bacterial enzymes)→ Short-chain fatty acids, gas

Fiber holds water; regulates bowel activity; and binds substances such as bile, cholesterol, and some minerals, carrying them out of the body.

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Absorption of carbohydrates

- The principal monosaccharides produced by the digestion of carbohydrates are glucose, fructose and galactose
- Glucose accounts for 80% of the total monosaccharides
- The absorption occurs mostly in the duodenum & upper jejunum of small intestine
- Only monosaccharides are absorbed by the intestine
- Absorption rate is maximum for galactose; moderate for glucose; and minimum for fructose

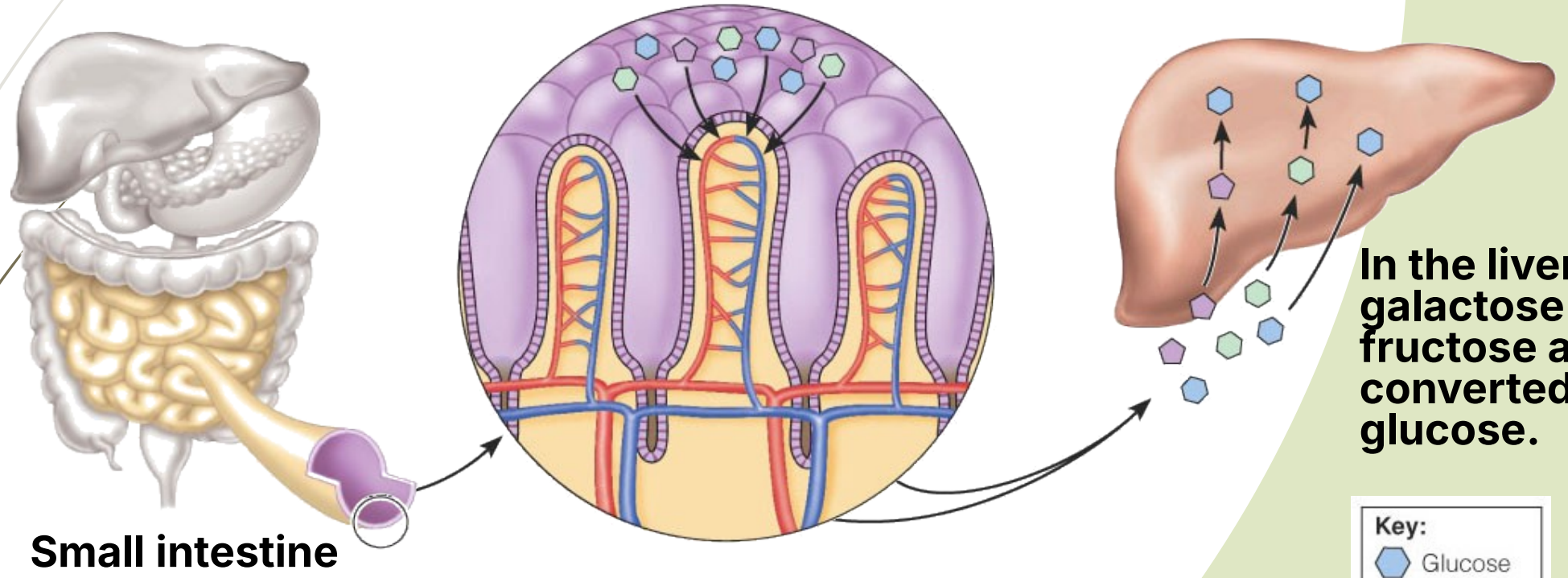
Absorption rates

- Cori study:
- He studies the rate of absorption of different sugars from small intestine in rat
- Glucose absorption as 100, comparative absorption of other sugars as
- Galactose=110, Glucose=100, Fructose=43, Mannoase=19, Xylose=15 & Arabinose=9
- Galactose is absorbed more rapidly than glucose
- Pentoses are absorbed slowly

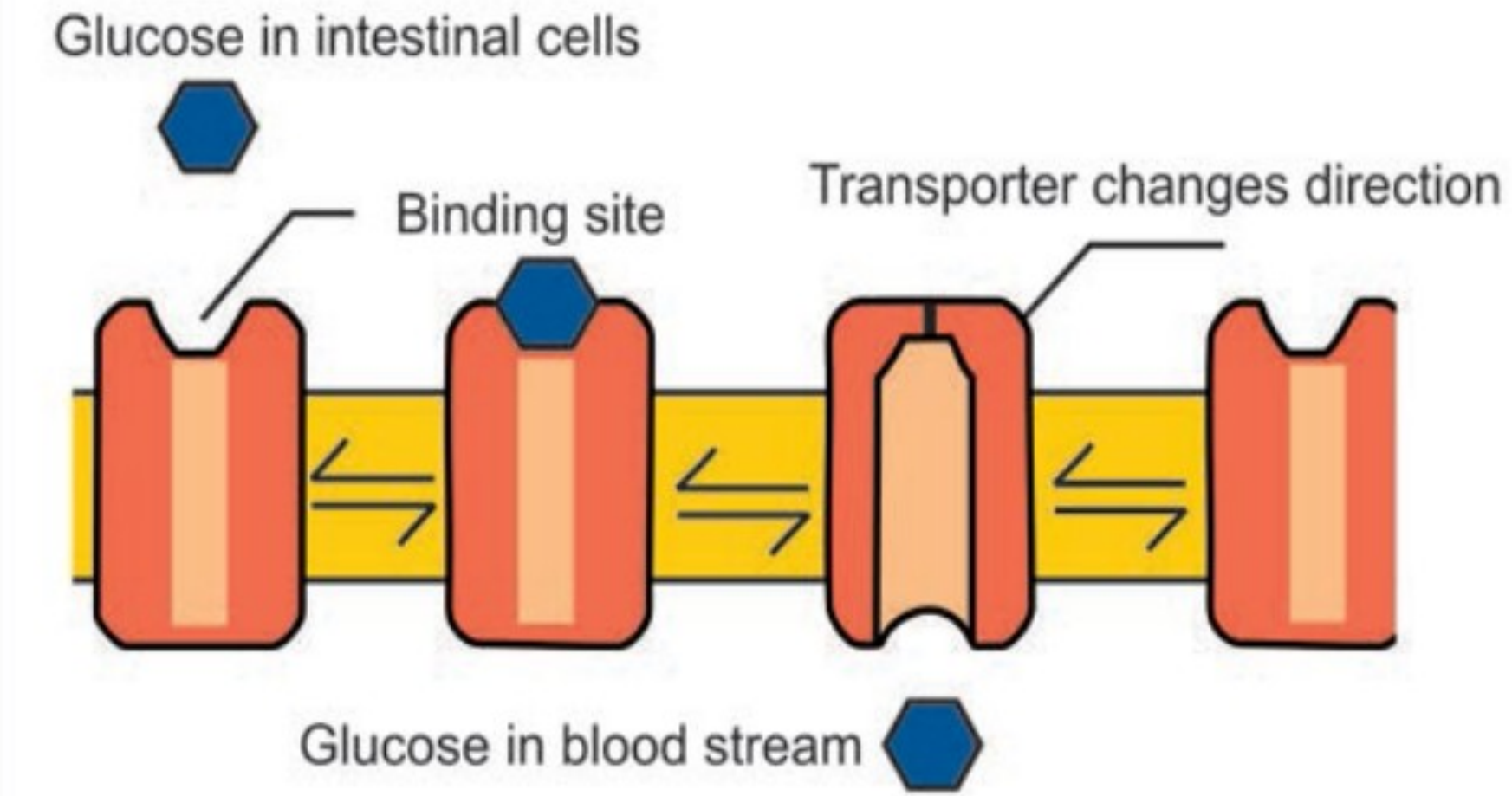
Mechanism of absorption

- Different sugars possess different mechanisms for their absorption
- Glucose is transported into the intestinal mucosal cells by a carrier mediated and energy requiring process

Monosaccharides, the end products of carbohydrate digestion, enter the capillaries of the intestinal villi



Glucose absorption (GluT-2)



Active transport mechanism

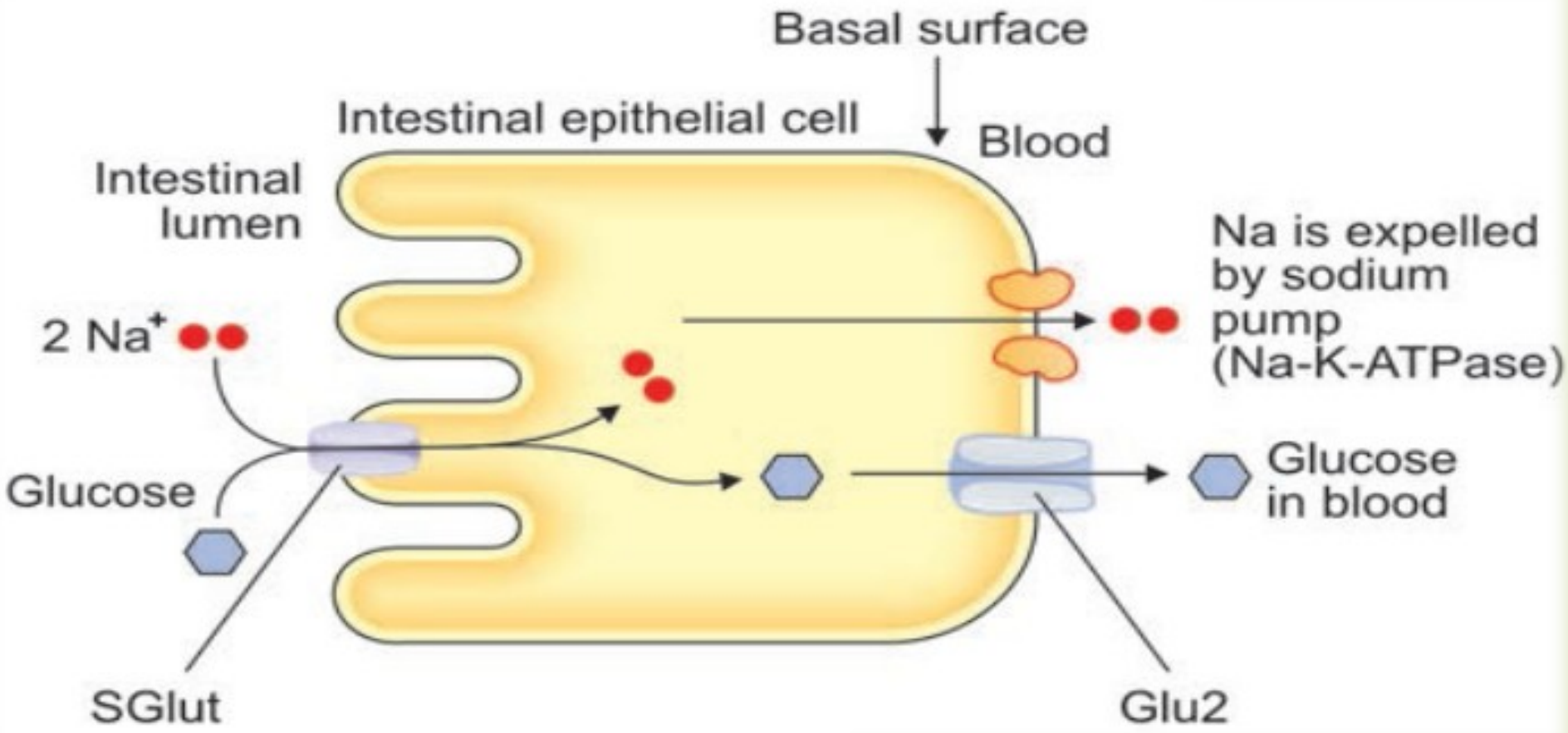
- Glucose and Na^+ share the same transport system (symport) referred to as sodium dependent glucose transporter
- The concentration of Na^+ is higher in the intestinal lumen compared to mucosal cells
- Na^+ moves into the cells along its concentration gradient & simultaneously glucose is transported into the intestinal cells
- Mediated by the same carrier system

Na^+ diffuses into the cell and it drags glucose along with it

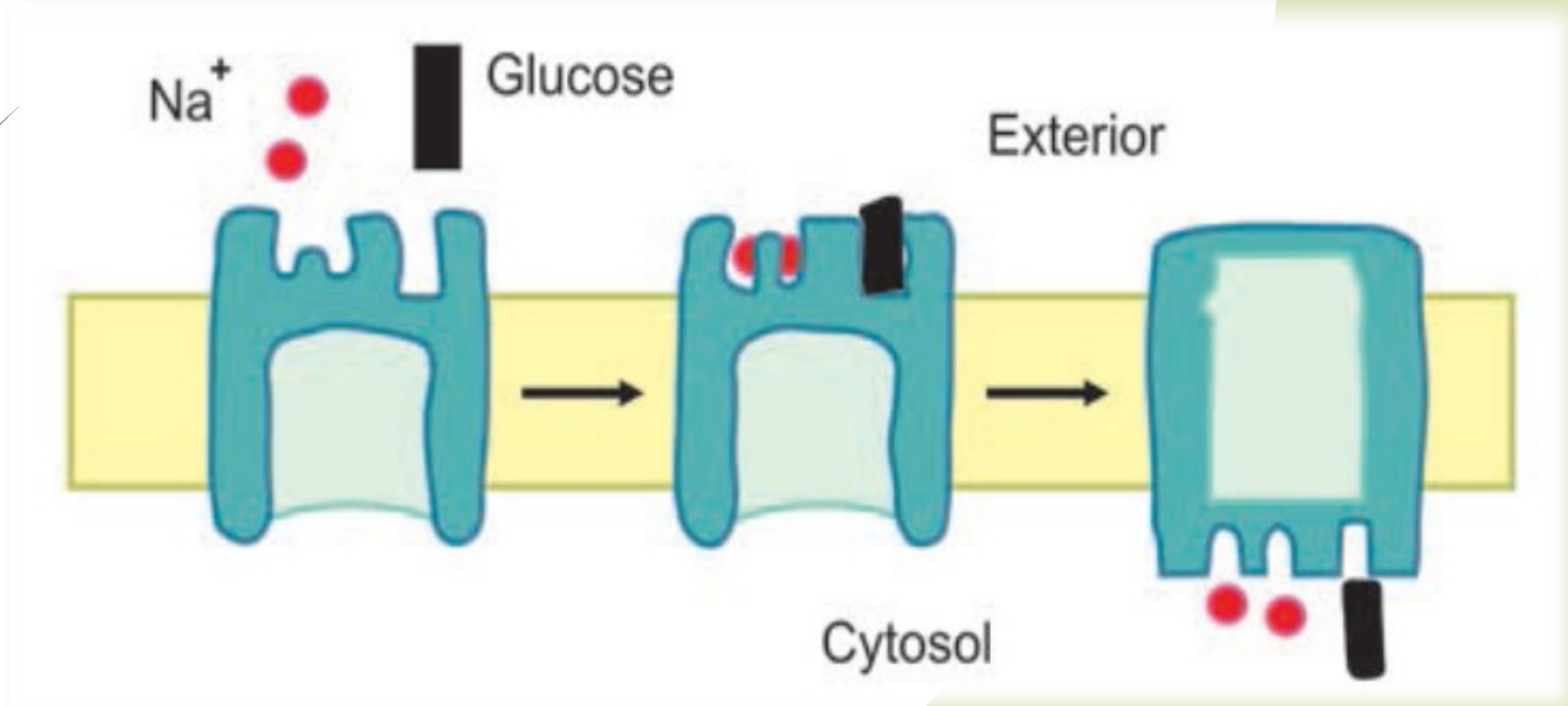
- The intestinal Na^+ gradient is the immediate energy source for glucose transport
- This energy is indirectly supplied by ATP since the re-entry of Na^+ (against the concentration gradient) into the intestinal lumen is an energy requiring active process
- The enzyme Na^+-K^+ ATPase is involved in the transport of Na^+ in exchange of K^+ against the concentration gradient

Intestinal absorption of glucose

- At the intestinal lumen, absorption is by SGluT & at the blood vessel side, absorption is by GluT2



SGluT: Sodium and glucose co-transport system at luminal side; sodium is then pumped out



Oral rehydration therapy (ORT):

- ORT is common treatment of diarrhoea
- Oral rehydration fluid contains glucose & sodium
- Intestinal absorption of sodium is facilitated by the presence of glucose
- Mechanism of absorption of galactose is similar to that of glucose
- Phlorozin blocks the Na^+ dependent transport of glucose & galactose

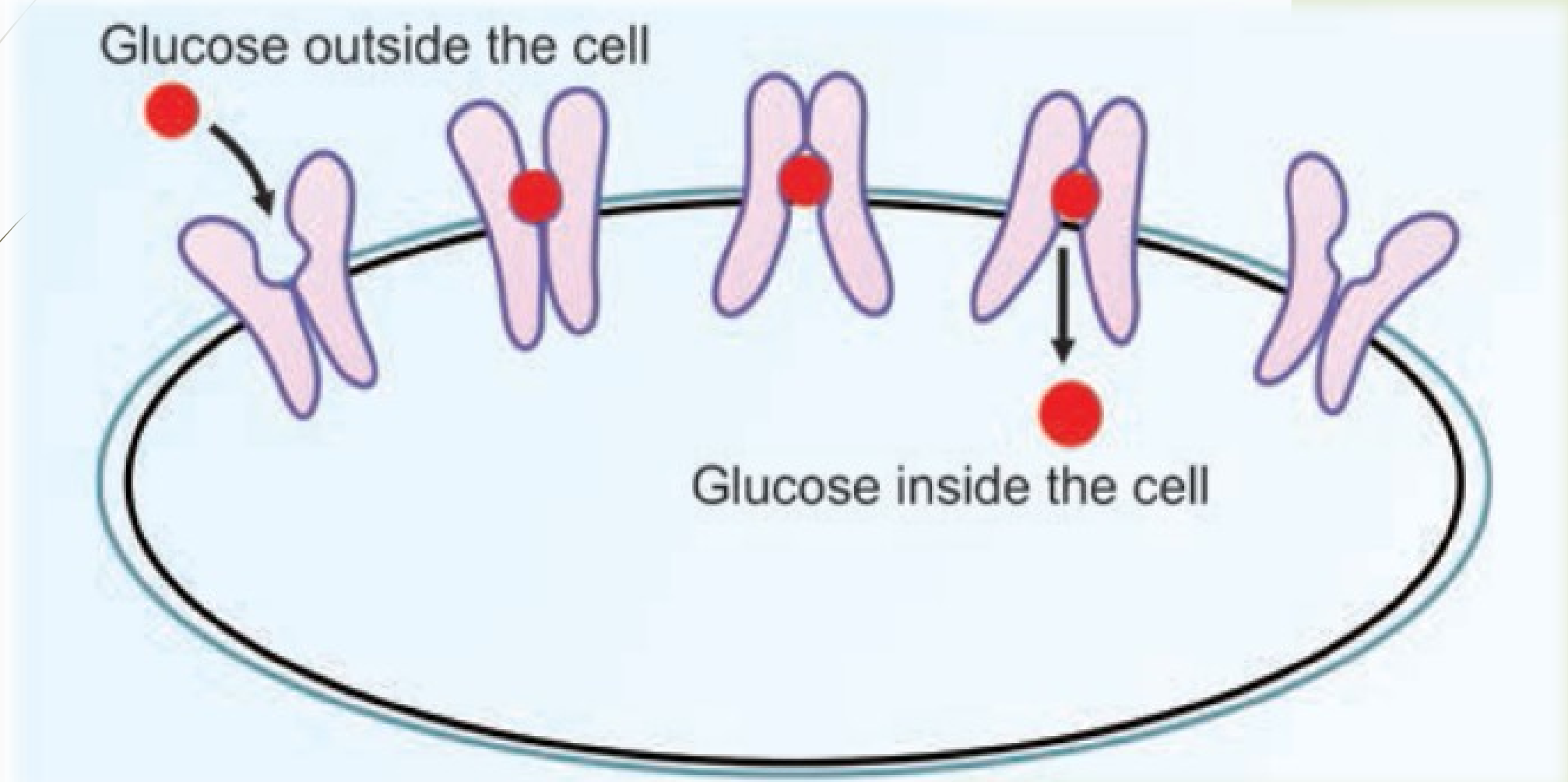
Glucose transporters

- Glucose transporters GluT-1 to 7 have been described in various tissues
- GluT-2 & GluT-4 are very important
- GluT-2:
- Operates in intestinal epithelial cells
- It is a uniport system & not dependent on Na^+ ions
- Glucose is held on GluT-2, by weak hydrogen bonds
- After fixing glucose, changes configuration & opens inner side releasing glucose

GluT-4:

- Operates in the muscle & adipose tissue
- GluT-4 is under control of insulin
- Insulin induces the intracellular GluT-4 molecules to move to the cell membrane & increases the uptake
- Other “GluT” molecules are ***not under control of insulin***
- GluT-1 is present in RBCs & brain
- Also present in retina, colon, placenta
- It helps in glucose uptake in most of these tissues which is independent of insulin

GluT4- Glucose transport in cells



Glucose transporters

Transporter	Present in	Properties
GluT1	RBC, brain, kidney, colon, retina, placenta	Glucose uptake in most of cells
GluT2	Surface of intestinal cells, liver, β-cells of pancreas	Low affinity; glucose uptake in liver; glucose sensor in β-cells
GluT3	Neurons, brain	High affinity; glucose into brain cells
GluT4	Skeletal, heart muscle, adipose tissue	Insulin mediated glucose uptake
GluT5	Small intestine, testis, sperms, kidney	Fructose transporter; poor ability to transport glucose
GluT7	Liver endoplasmic reticulum	Glucose from ER to cytoplasm
SGluT	Intestine, kidney	Cotransport; from lumen into cell

Absorption of fructose:

- Fructose absorption is simple
- Does not require energy and Na^+ ions
- Transported by facilitated diffusion mediated by a carrier
- Inside the epithelial cell, most of the fructose is converted to glucose
- The latter then enters the circulation
- Pentoses are absorbed by a process of simple diffusion

Factors influencing rate of absorption

- Mucus membrane:
- Mucus membrane is not healthy, absorption will decrease
- Thyroid hormones:
- Increases absorption of hexoses & act on intestinal mucosa
- Adrenal cortex: Absorption decreases in adrenocortical deficiency, mainly due to decreased concentration of sodium
- Anterior pituitary: It affects mainly through thyroid hormones

- Insulin:
- It has no effect on absorption of glucose
- Vitamins:
- Absorption is decreased in B-complex vitamins deficiency-thiamine, pyridoxine, pantothenic acid
- Inherited deficiency of sucrase & lactase enzymes interfere with corresponding disaccharide absorption

Abnormalities of carbohydrate digestions

- Defect in disaccharidases results in the passage of undigested disaccharides into the large intestine
- The disaccharides draw water from the intestinal mucosa by osmosis and cause osmotic diarrhoea
- Bacterial action of these undigested carbohydrates leads to flatulence
- Flatulence is characterized by increased intestinal motility, cramps and irritation

- The carbohydrates (di, oligo and polysaccharides) not hydrolysed by α-amylase
- The di & oligosaccharides can be degraded by the bacteria present in ileum to liberate monosaccharides
- During the course of utilization of monosaccharides by the intestinal bacteria, the gases such as hydrogen, methane & carbon dioxide-besides lactate and short chain fatty acids are released & causes flatulence

- The occurrence of flatulence after the ingestion of leguminous seeds (bengal gram, redgram, beans, peas, soya bean) is very common
- They contain several non-digestible oligonccharides by human intestinal enzymes
- These compounds are degraded and utilised by intestinal bacteria causing flatulence
- Raffinose containing galactose, glucose and fructose is a predominant oligosaccharide found in leguminous seeds

Lactose intolerance

- lactase (β-galactosidase) deficiency is the most common disaccharidase deficiency in humans
- It is estimated that more than half of the world's adult population is affected by lactose intolerance
- Some infants may have deficiency of lactase & they show intolerance to lactose, the milk sugar
- Symptoms:
- Diarrhoea, flatulence, abdominal cramps

Discussion:

- Lactose of milk cannot be hydrolysed due to deficiency of lactase
- Accumulation of lactose in intestinal tract, which is “osmotically active” & holds water, producing diarrhoea.
- Accumulated lactose is also fermented by intestinal bacteria which produce gas & other products, producing flatulence & abdominal pain

Sucrase deficiency:

- Inherited deficiency of sucrose
- Symptoms occurs in early childhood with ingestion of sugars, sucrose
- Symptoms: Diarrhoea, flatulence, abdominal cramps
- Disaccharidiuria:
- Increase in the excretion of disaccharides may be observed in some patients with disaccharidase deficiency
- Observed in intestinal damage, celiac diseases

Thank you